

The Earliest Deadline Late (EDL) Server

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Source: G. C. Buttazzo, Hard Real-Time Computing Systems, 3rd ed., Section 6.5

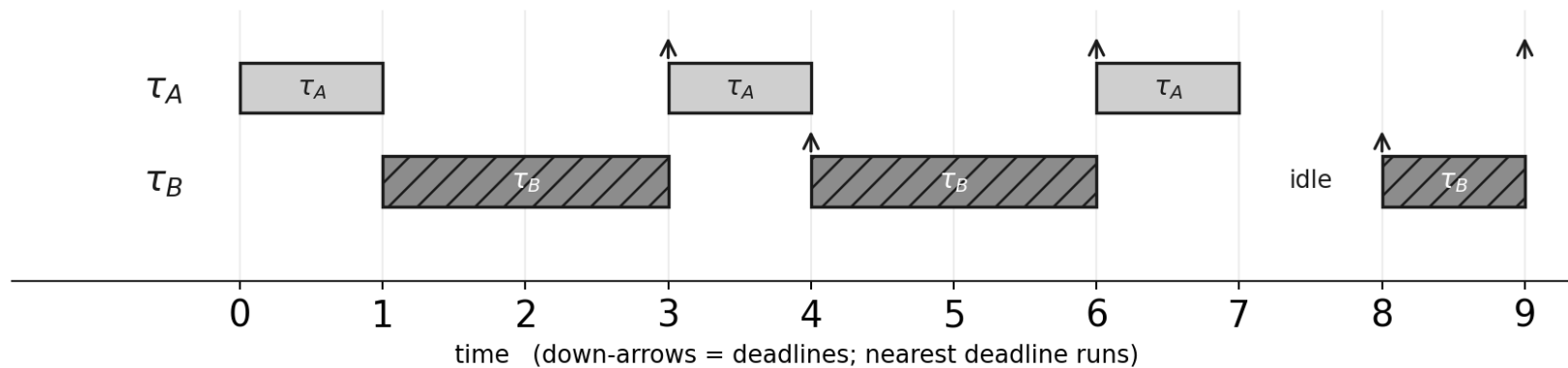
Two Workloads, One Processor

A real-time result must be correct and on time.

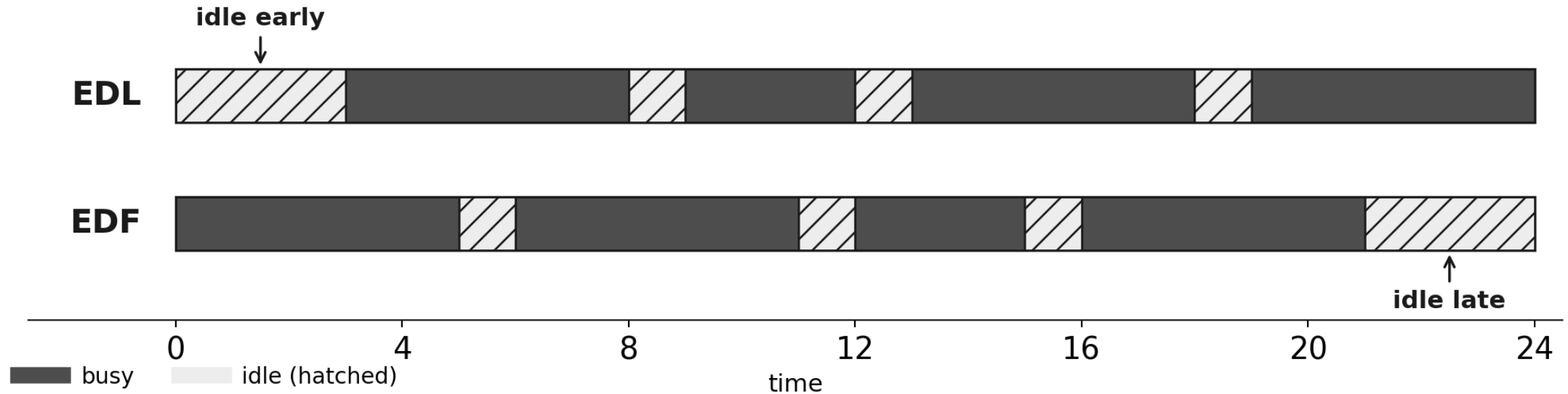
- Periodic tasks — fixed period, hard deadlines that must never be missed
- Aperiodic requests — arrive unpredictably, should be served quickly
- Background service is safe but slow; immediate service is fast but unsafe
- **Goal: an aperiodic server — fast and provably safe**

Earliest Deadline First (EDF)

- Run the ready job with the nearest absolute deadline
- Dynamic priority; optimal on one processor
- Feasible iff $U_p = \sum C_i / T_i \leq 1$ (implicit deadlines, $D = T$)

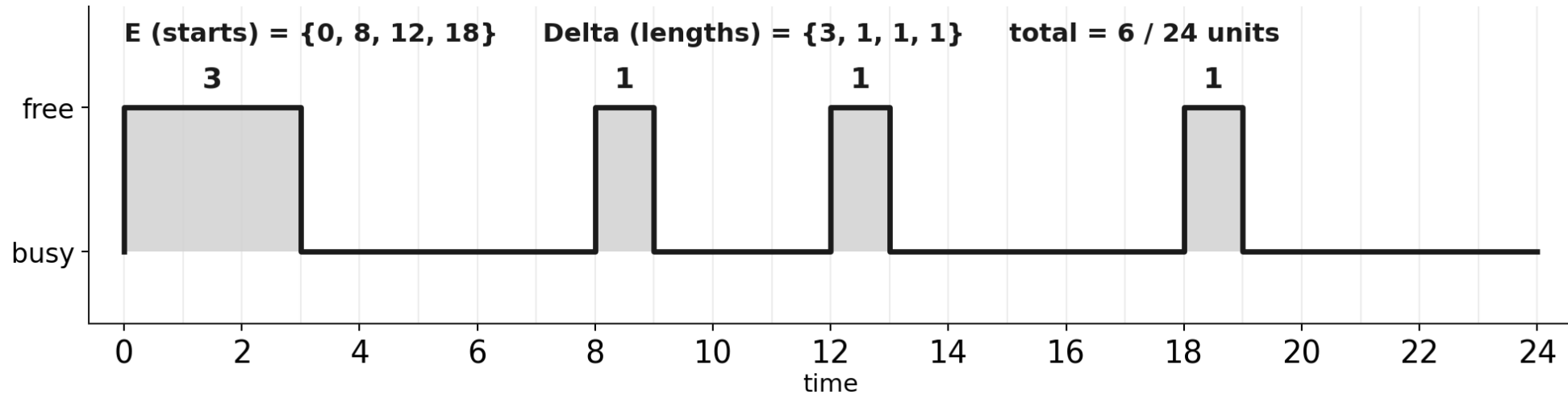


EDF vs. EDL: Free the Processor Early



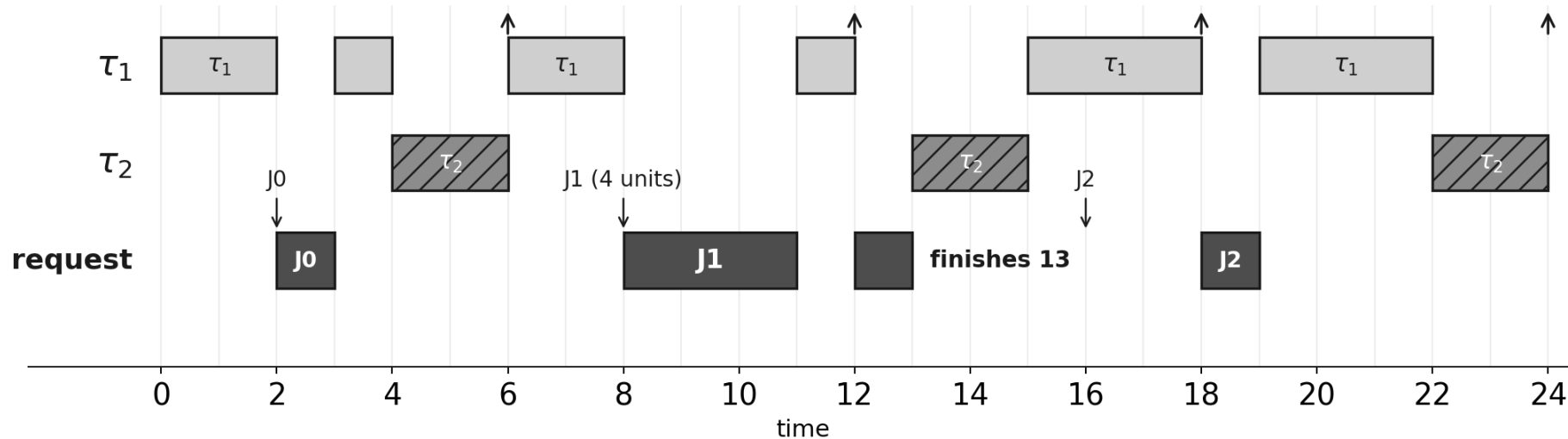
- EDF — as soon as possible: idle time accumulates late
- EDL — as late as safely possible: idle time appears early
- **Theorem (Chetto & Chetto):** EDL leaves the maximum idle time in every $[0, t]$

The EDL Server and Its Idle-Time Map



- No request $\rightarrow$ run EDF; request $\rightarrow$ fill the earliest idle slots first
- Gets the spare bandwidth $U_s = 1 - U_p$ automatically (here 25%)
- **A slot is usable when slack $S(t) = \min \text{ over } d \text{ of } [(d - t) - g(t, d)] \geq 1$**

Worked Example: Request at $t = 8$



- $\tau_1=(3,6)$, $\tau_2=(2,8)$: $U_p = 0.75$, spare bandwidth 25%, $H = 24$
- Request at $t = 8$ (4 units): slack $S(8) = \min\{3,5,4,5\} = 3 \rightarrow$ runs at 8, 9, 10
- **τ_1 forced at 11; last unit at 12 $\rightarrow$ finishes at 13, response $R = 5$ (optimal)**

Correctness: Schedulable and Optimal

Safe — schedulability

Requests use only would-be-idle time, so the periodic guarantee is unchanged:

$$\text{schedulable} \Leftrightarrow U_p \leq 1$$

Optimal — response time

No on-line algorithm finishes a request sooner than the EDL server:

$$f_{EDL}(J) \leq f_A(J)$$

Cost, Practical Form

- Cost: rebuild the idle map per arrival $\rightarrow O(N \cdot n)$ (N = instances per hyperperiod)
- Assumes known periodic parameters, known request WCETs, and $D = T$
- IPE server — precompute the map off-line $\rightarrow$ near-optimal at $O(1)$ per event

Strengths, Limitations, Applications

Strengths

- Provably optimal response
- Preserves the bound $U_p \leq 1$
- Uses spare bandwidth automatically

Limitations

- $O(N \cdot n)$ on-line cost
- Needs each request's WCET
- Fixed, known task set

Applications

- Multimedia + UI events
- Network servers (bursty load)
- Robotics, industrial control

The EDL Server in One Picture



Optimal · Safe whenever $U_p \leq 1$ · Practical via the IPE server

Schedule the chores late — serve the requests early.